

Gioia Zheng

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Research Profile

Applied Computer Science and AI undergraduate at Sapienza University of Rome working on safety-relevant evaluation of RAG and LLM systems. I build reproducible pipelines for paired comparisons, grounding checks, trace-level diagnostics, and failure taxonomies that expose regressions hidden by aggregate metrics.

Education

Sapienza University of Rome

Sep 2022 – Present

B.Sc. in Applied Computer Science and Artificial Intelligence

Research Experience

NLP Algorithm Engineer Intern, Microsoft (China)

Feb 2026 – May 2026

Natural Language Computing, remote

- Designed and implemented a multi-stage retrieval and generation evaluation pipeline over MS MARCO-scale data: BM25 retrieval, FAISS dense retrieval, transformer reranking, and seq2seq generation.
- Built paired evaluation routines with confidence intervals to measure how stage-level retrieval changes propagate to answer quality and grounding behavior.
- Defined schema-versioned experiment manifests and validation checks so results remain traceable to code, data, configuration, and environment.

Selected Research Projects

MS MARCO Generative QA System

github.com/GioiaZheng/msmarco-genqa

Python, PyTorch, BM25, FAISS, cross-encoder reranking, T5, grounding evaluation

- Study when retrieval and reranking gains translate into faithful, evidence-supported generation rather than only higher aggregate surface metrics.
- Built a reproducible retrieve → rerank → generate pipeline on MS MARCO passage data with config-driven experiments and schema-versioned manifests.
- Reported Token-F1 improvement from 0.197 to 0.368 on 6,980 paired queries, with paired-bootstrap confidence interval [+0.163, +0.178].
- Developed a failure taxonomy separating retrieval, reranking, evidence-use, and generation errors to expose regressions hidden by aggregate scores.

RAG Observatory

github.com/GioiaZheng/rag-observatory

Python, trace schemas, failure taxonomy, evidence attribution, diagnostic reports

- Building a trace-based observability toolkit for RAG runs covering queries, retrieved candidates, context, answers, evidence, evaluation signals, and failure labels.
- Implemented schema validation, failure-label checks, diagnostic reports, and comparison workflows for inspecting support, abstention behavior, and metric-hidden regressions.

Additional Experience

Web Development Intern, Idearia

Mar 2026 – Jun 2026

Maintain production systems and automate operational workflows in a client-facing environment.

Young Representative, UPI Emilia-Romagna

Sep 2022 – Oct 2023

Conducted data-driven research on school dropout and contributed to policy-oriented reports.

Skills

Programming

Python, SQL

ML and Retrieval

Information retrieval, RAG and LLM evaluation, PyTorch, FAISS, BM25, Sentence Transformers, cross-encoder reranking, T5

Research Methods

Paired bootstrap, confidence intervals, controlled ablations, grounding evaluation, failure taxonomy, trace-level diagnostics

Research Tooling Languages

Linux, Docker, Git, GitHub Actions, experiment manifests, reproducible pipelines
Chinese (Native), Italian (Bilingual), English (Professional), German (Beginner)